

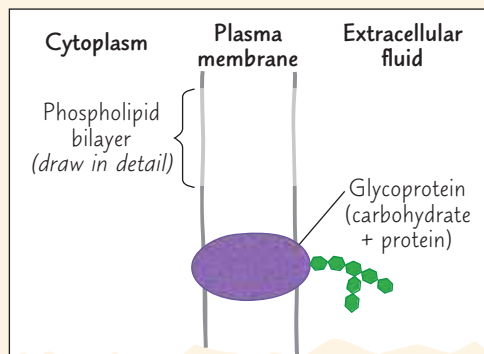
7 Membrane Structure and Function

KEY CONCEPTS

- 7.1** Cellular membranes are fluid mosaics of lipids and proteins *p. 127*
- 7.2** Membrane structure results in selective permeability *p. 131*
- 7.3** Passive transport is diffusion of a substance across a membrane with no energy investment *p. 132*
- 7.4** Active transport uses energy to move solutes against their gradients *p. 136*
- 7.5** Bulk transport across the plasma membrane occurs by exocytosis and endocytosis *p. 139*

Study Tip

Make a visual study guide: Draw a plasma membrane (two lines) all the way down a piece of paper (or digitally). Label the cytoplasm and the extracellular fluid. At the top, draw the phospholipid bilayer in detail. As you read the chapter, draw and label membrane proteins that you encounter and diagram the different ways materials can enter or leave a cell.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 7
- Figure 7.12 Walkthrough: Osmosis
- BioFlix® Animation: Membrane Transport

For Instructors (in Item Library)

- Tutorial: Membrane Transport: Diffusion and Passive Transport
- Tutorial: Membrane Transport: Bulk Transport



Figure 7.1 Successful learning relies on communication between brain cells. Here, the vesicles fusing with the plasma membrane of the top cell release molecules (yellow) that bind to membrane proteins (light green) on the surface of the bottom cell, triggering the proteins to change shape. The plasma membrane that surrounds each cell regulates its exchanges with its environment and surrounding cells.

How does the plasma membrane regulate inbound and outbound traffic?

